# Caching

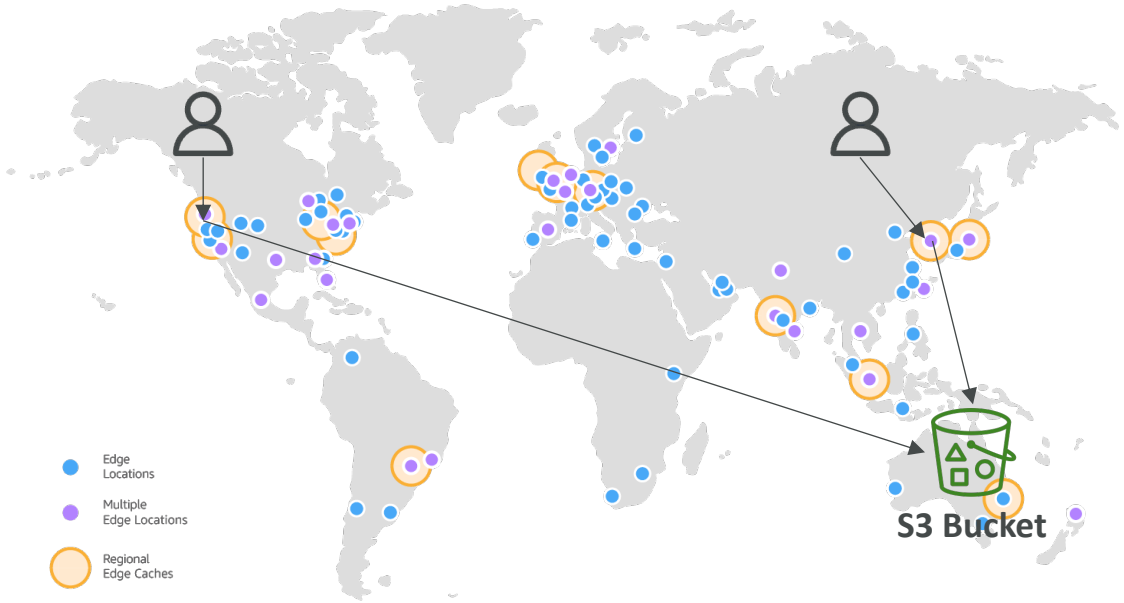
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# Amazon CloudFront



- Content Delivery Network (CDN)
- Improves read performance, content is cached at the edge
- 225+ Point of Presence globally (215+ Edge Locations & 13 Regional Edge Caches)
- Protect against Network and Application layer attacks (e.g., DDoS attacks)
- Integration with AWS Shield, AWS WAF, and Route 53
- Can expose external HTTPS and can talk to internal HTTPS backends
- Supports WebSocket protocol




Source: https://aws.amazon.com/cloudfront/features/

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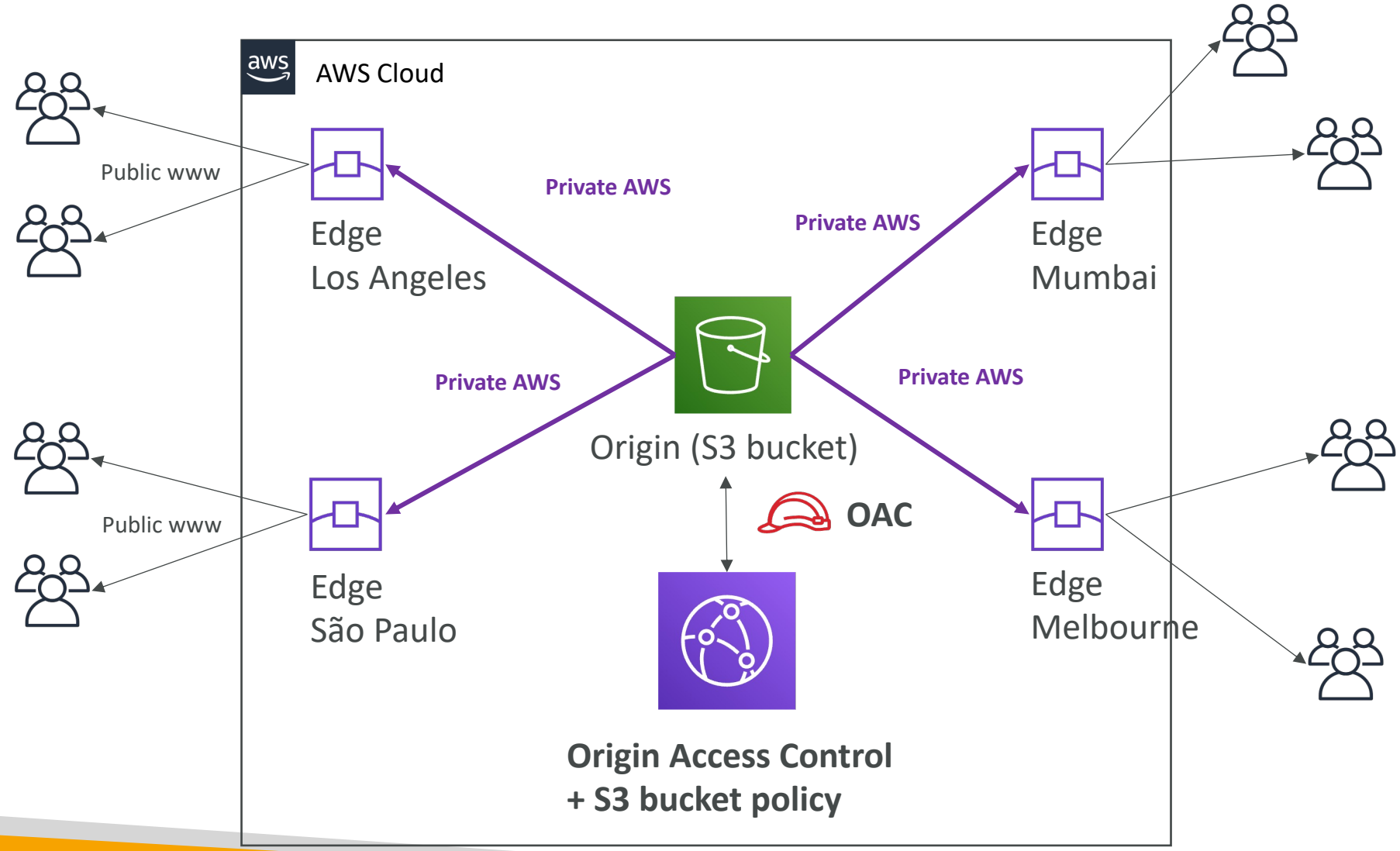
# CloudFront – Origins

- S3 Bucket
  - For distributing files
  - For uploading files to S3 (using CloudFront as an ingres)
  - Enhanced security with CloudFront **Origin Access Control (OAC)**
- MediaStore Container & MediaPackage Endpoint
  - To deliver Video On Demand (VOD) or live streaming video using AWS Media Services
- VPC Origin
  - For applications hosted in VPC private subnets
  - Application Load Balancer / Network Load Balancer / EC2 Instances
- Custom Origin (HTTP)
  - API Gateway (for more control… otherwise use API Gateway Edge)
  - S3 Bucket configured as a website (enable Static Website hosting)
  - Any HTTP backend you want

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# CloudFront – S3 as an Origin




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# CloudFront vs S3 Cross Region Replication

- CloudFront:
  - Global Edge network
  - Files are cached for a TTL (maybe a day)
  - **Great for static content that must be available everywhere**

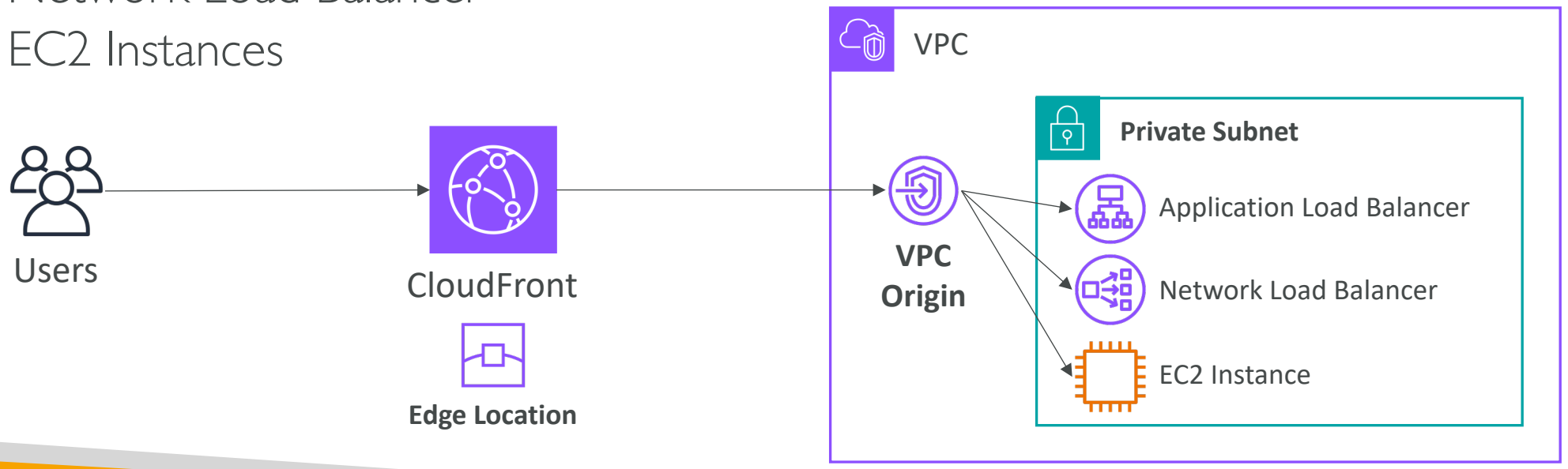
- S3 Cross Region Replication:
  - Must be setup for each region you want replication to happen
  - Files are updated in near real-time
  - Read only
  - **Great for dynamic content that needs to be available at low-latency in few regions**

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# CloudFront – ALB or EC2 as an origin Using VPC Origins

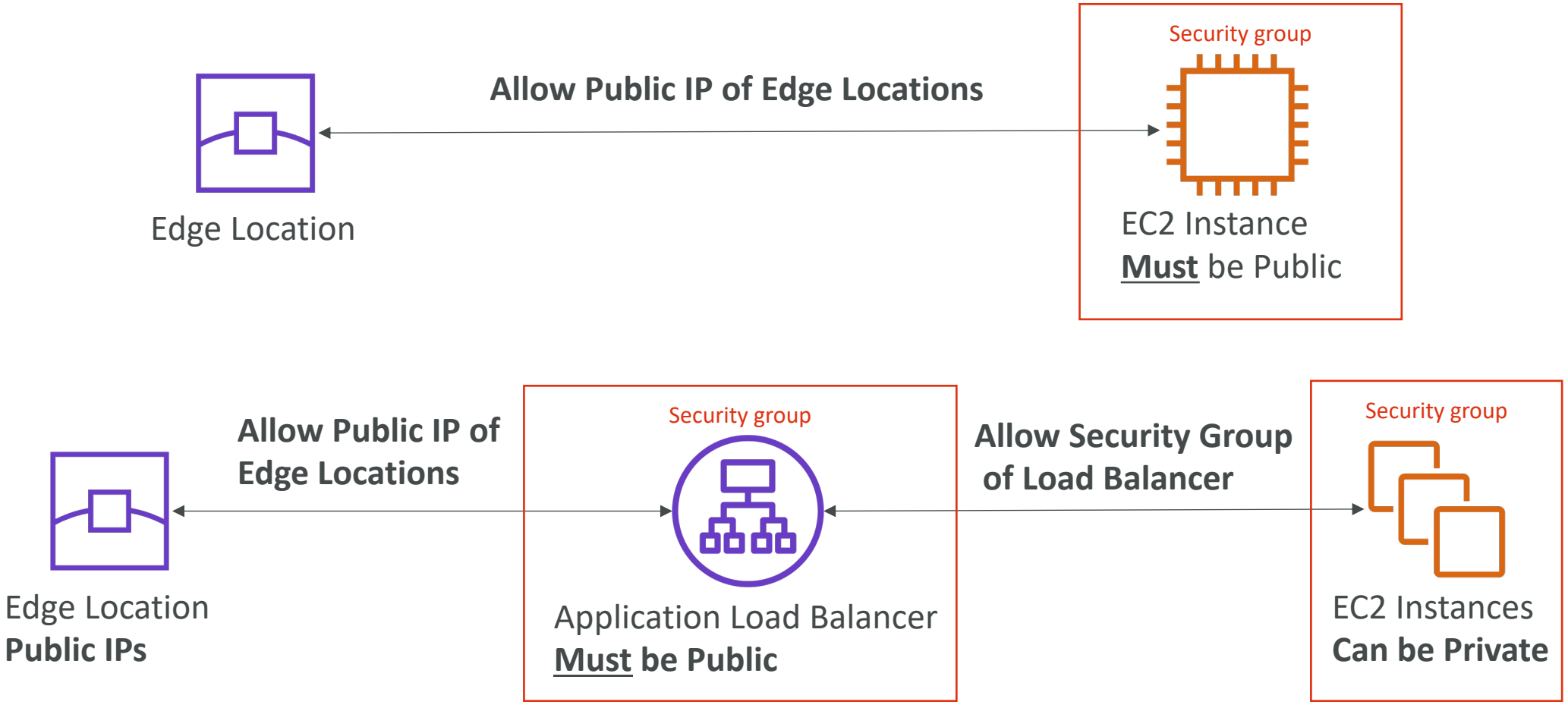
- Allows you to deliver content from your applications hosted in your VPC private subnets (no need to expose them on the Internet)
- Deliver traffic to **private**:
  - Application Load Balancer
  - Network Load Balancer
  - EC2 Instances




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# CloudFront – EC2 or ALB as an origin



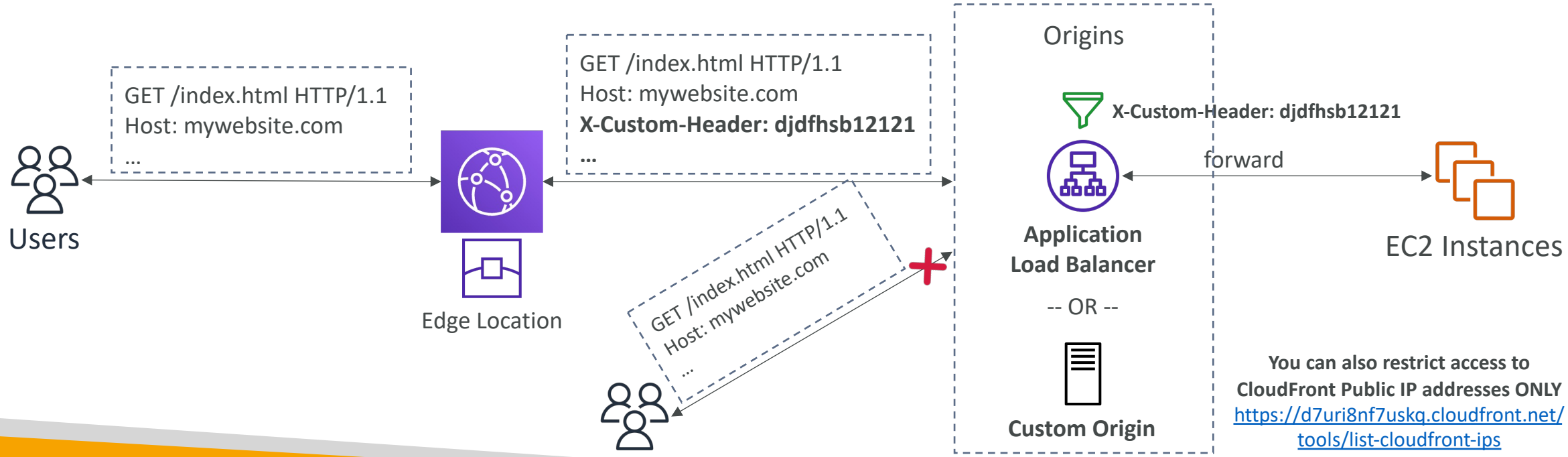

http://d7uri8nf7uskq.cloudfront.net/tools/list-cloudfront-ips

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# CloudFront – Restrict Access to Application Load Balancers and Custom Origins

- Prevent direct access to your ALB or Custom Origins (only access through CloudFront)
- First, configure CloudFront to add a **Custom HTTP Header** to requests it sends to the ALB
- Second, configure the ALB to only forward requests that contain that Custom HTTP Header
- Keep the custom header name and value secret!

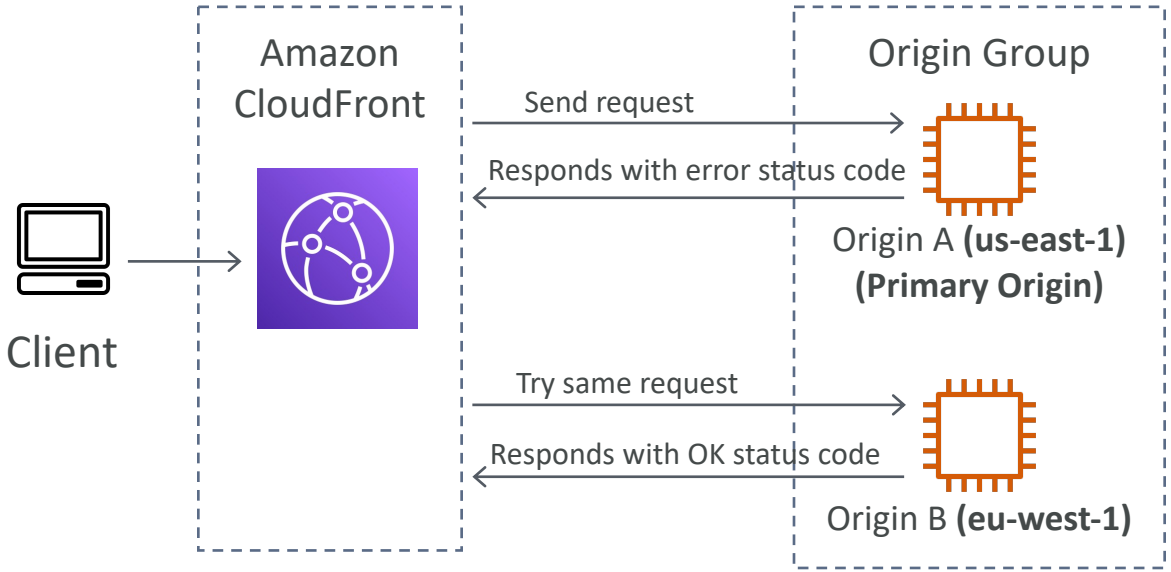



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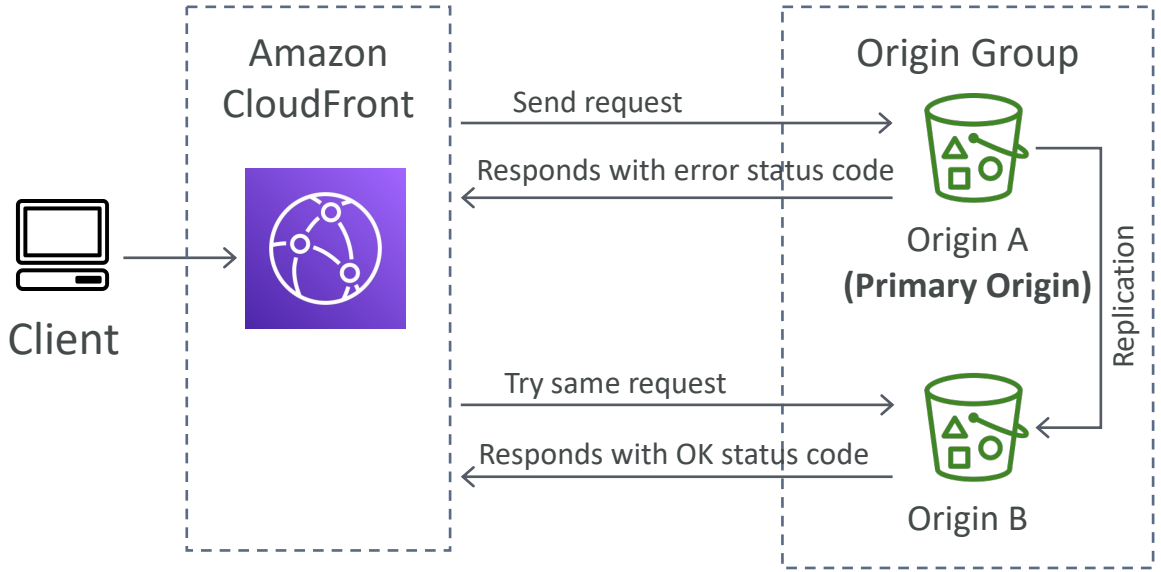
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# CloudFront – Origin Groups

- To increase high-availability and do failover
- Origin Group: one primary and one secondary origin
- If the primary origin fails, the second one is used
- Origins can be cross AWS Regions




Cross-Region High Availability




S3 + CloudFront – Region-level High Availability

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# CloudFront Geo Restriction

- You can restrict who can access your distribution
  - **Allow list:** Allow your users to access your content only if they're in one of the countries on a list of approved countries.
  - **Block list:** Prevent your users from accessing your content if they're in one of the countries on a blacklist of banned countries.
- The "country" is determined using a 3rd party Geo-IP database
- Use case: Copyright Laws to control access to content
- Note: the geo header *CloudFront-Viewer-Country* is in Lambda@Edge

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# CloudFront – Pricing

- CloudFront Edge locations are all around the world
- The cost of data out per edge location varies

| Per Month | United States, Mexico, & Canada | Europe & Israel | South Africa, Kenya, & Middle East | South America | Japan | Australia & New Zealand | Hong Kong, Philippines, Singapore, South Korea, Taiwan, & Thailand | India |
|---|---|---|---|---|---|---|---|---|
| First 10TB | $0.085 | $0.085 | $0.110 | $0.110 | $0.114 | $0.114 | $0.140 | $0.170 |
| Next 40TB | $0.080 | $0.080 | $0.105 | $0.105 | $0.089 | $0.098 | $0.135 | $0.130 |
| Next 100TB | $0.060 | $0.060 | $0.090 | $0.090 | $0.086 | $0.094 | $0.120 | $0.110 |
| Next 350TB | $0.040 | $0.040 | $0.080 | $0.080 | $0.084 | $0.092 | $0.100 | $0.100 |
| Next 524TB | $0.030 | $0.030 | $0.060 | $0.060 | $0.080 | $0.090 | $0.080 | $0.100 |
| Next 4PB | $0.025 | $0.025 | $0.050 | $0.050 | $0.070 | $0.085 | $0.070 | $0.100 |
| Over 5PB | $0.020 | $0.020 | $0.040 | $0.040 | $0.060 | $0.080 | $0.060 | $0.100 |




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# CloudFront – Price Classes

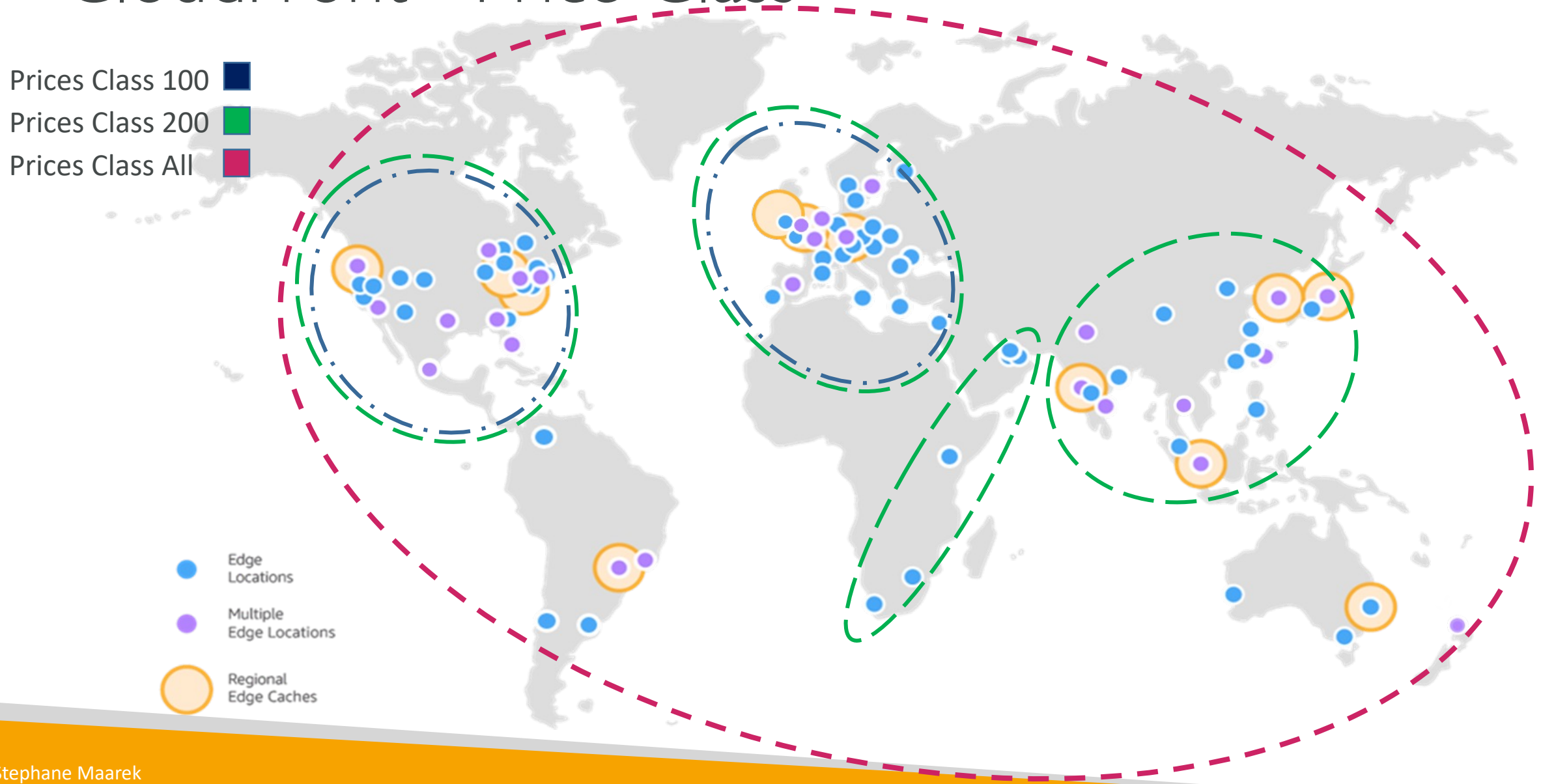
- You can reduce the number of edge locations for **cost reduction**
- Three price classes:
  1. Price Class All: all regions – best performance
  2. Price Class 200: most regions, but excludes the most expensive regions
  3. Price Class 100: only the least expensive regions

| Edge Locations Included Within | **United States, Mexico, & Canada** | **Europe & Israel** | **South Africa, Kenya, & Middle East** | **South America** | **Japan** | **Australia & New Zealand** | **Hong Kong, Philippines, Singapore, South Korea, Taiwan, & Thailand** | **India** |
|---|---|---|---|---|---|---|---|---|
| Price Class All | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Price Class 200 | Yes | Yes | Yes | x | Yes | x | Yes | Yes |
| Price Class 100 | Yes | Yes | x | x | x | x | x | x |

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# CloudFront - Price Class

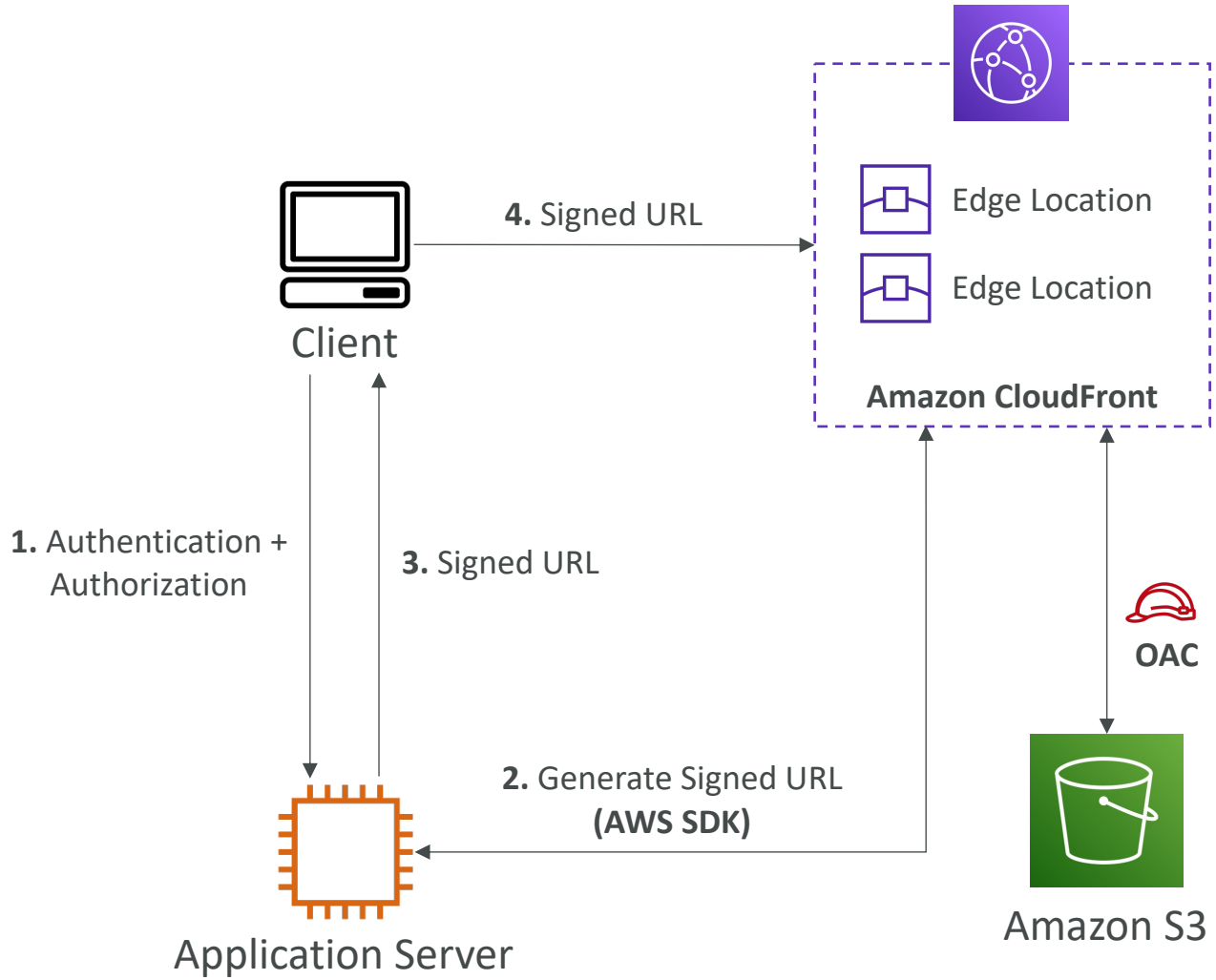



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# CloudFront Signed URL Diagram

- Signed URL with expiration to control access to content in CloudFront
- The Signed URL are generated by an API call into CloudFront as a trusted signer

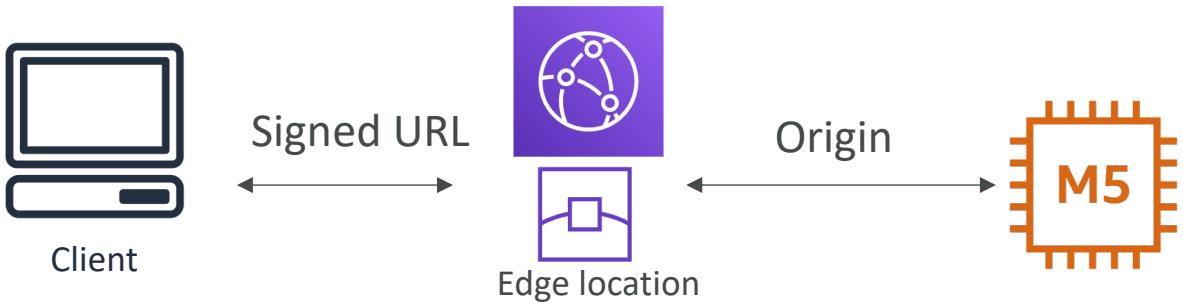


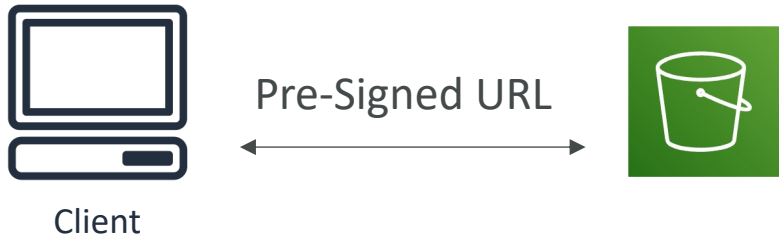

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# CloudFront Signed URL vs S3 Pre-Signed URL

- **CloudFront Signed URL:**
  - Allow access to a path, no matter the origin
  - Account wide key-pair, only the root can manage it
  - Can filter by IP, path, date, expiration
  - Can leverage caching features
- **S3 Pre-Signed URL:**
  - Issue a request as the person who pre-signed the URL
  - Uses the IAM key of the signing IAM principal
  - Limited lifetime



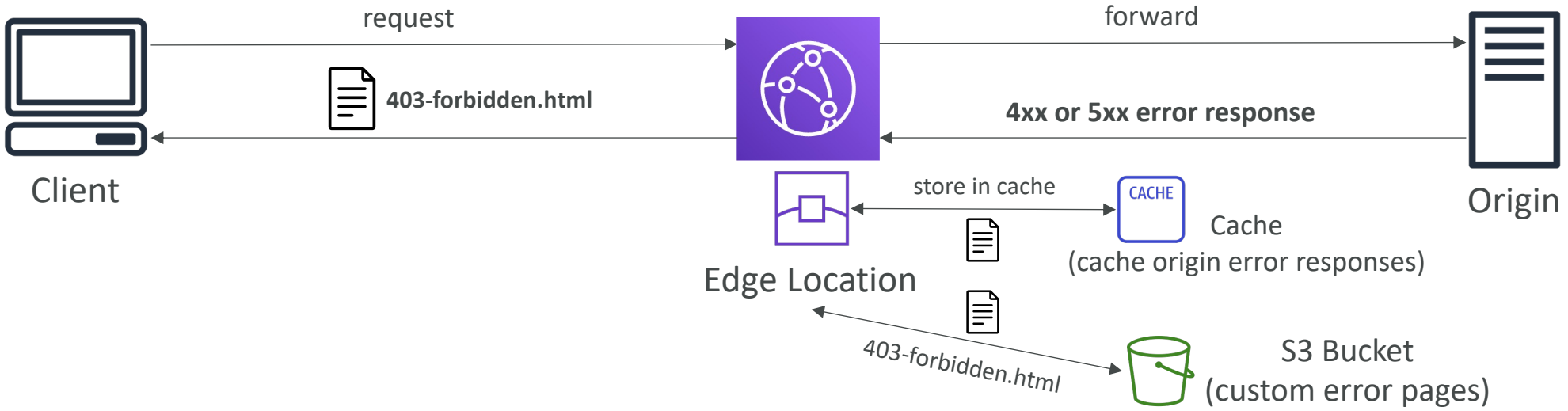




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# CloudFront – Custom Error Pages

- Return an object to the viewer (e.g., .html) when your origin returns an HTTP 4xx or 5xx status code to CloudFront
- Use **Error Caching Minimum TTL** to specify how long CloudFront caches the custom error pages




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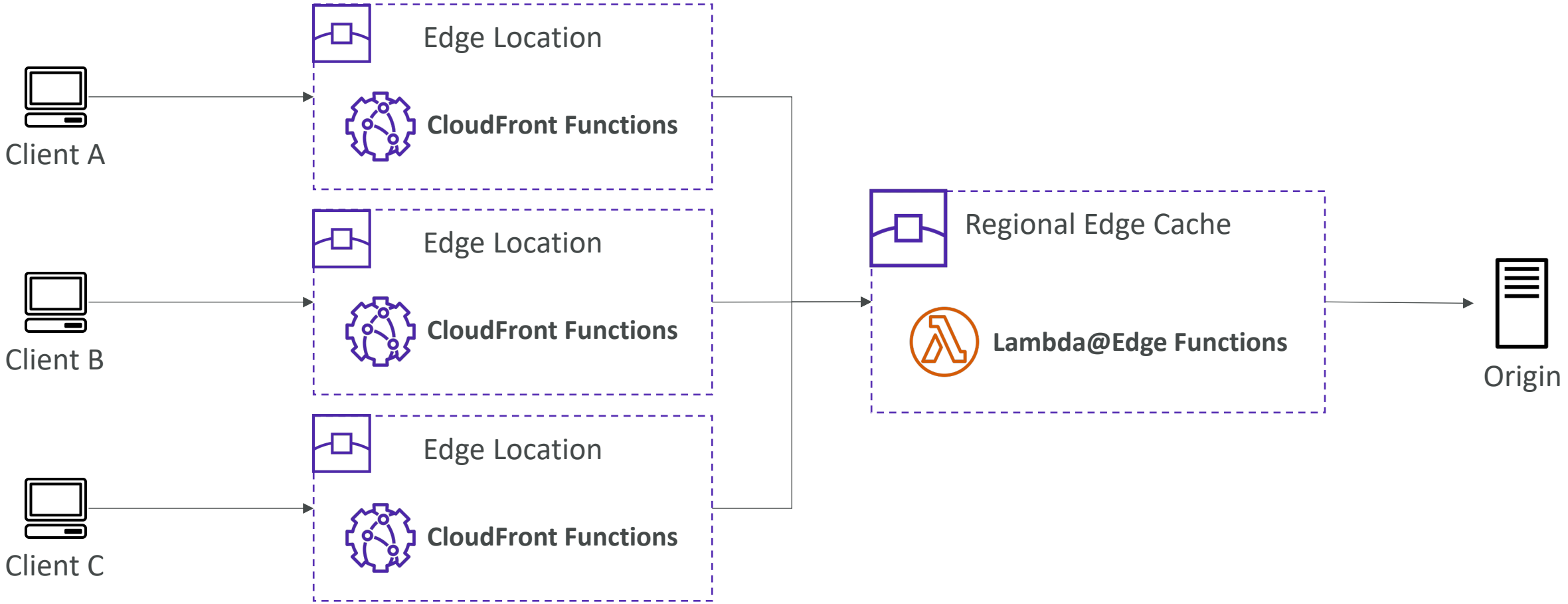
# CloudFront – Customization At The Edge

- Many modern applications execute some form of the logic at the edge
- **Edge Function:**
  - A code that you write and attach to CloudFront distributions
  - Runs close to your users to minimize latency
  - Doesn't have any cache, only to change requests/responses
  - CloudFront provides two types: **CloudFront Functions** & **Lambda@Edge**
- Use cases:
  - Manipulate HTTP requests and responses
  - Implement request filtering before reaching your application
  - User authentication and authorization
  - Generate HTTP responses at the edge
  - A/B Testing
  - Bot mitigation at the edge
- You don't have to manage any servers, deployed globally

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# CloudFront Functions & Lambda@Edge

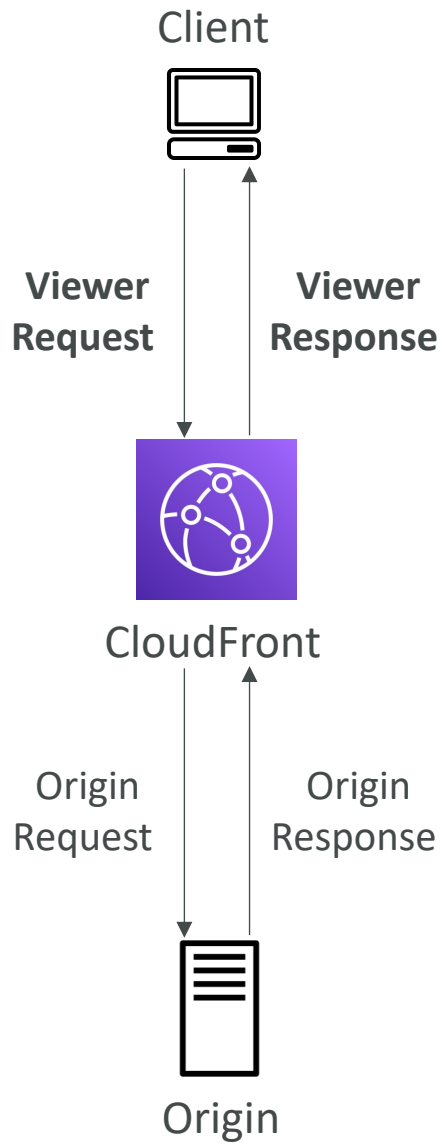



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# CloudFront – CloudFront Functions

- Lightweight functions written in JavaScript
- For high-scale, latency-sensitive CDN customizations
- Sub-ms startup times, millions of requests/second
- RuN at Edge Locations
- Process-based isolation
- Used to change Viewer requests and responses:
  - **Viewer Request:** after CloudFront receives a request from a viewer
  - **Viewer Response:** before CloudFront forwards the response to the viewer
- Native feature of CloudFront (manage code entirely within CloudFront)

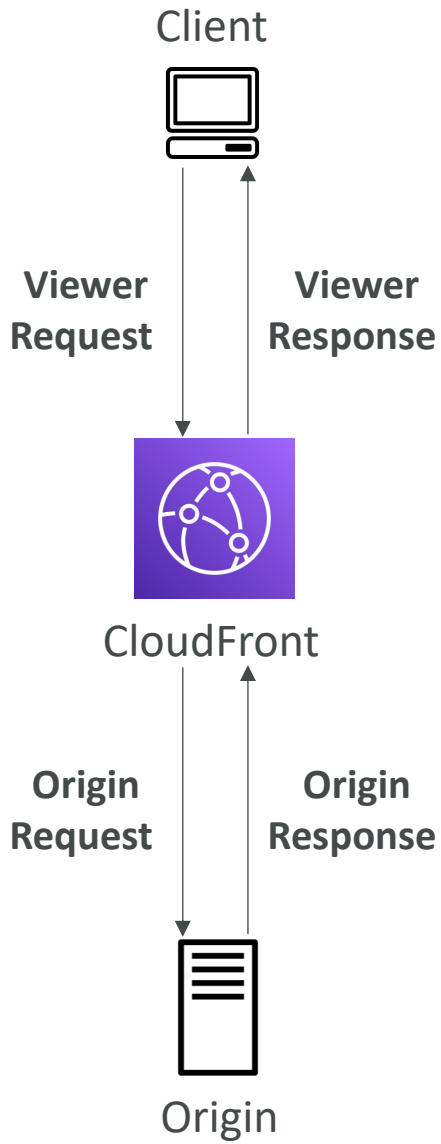



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# CloudFront – Lambda@Edge

- Lambda functions written in NodeJS or Python
- Scales to 1000s of requests/second
- Runs at the nearest Regional Edge Cache
- VM-based isolation
- Used to change CloudFront requests and responses:
  - **Viewer Request** – after CloudFront receives a request from a viewer
  - **Origin Request** – before CloudFront forwards the request to the origin
  - **Origin Response** – after CloudFront receives the response from the origin
  - **Viewer Response** – before CloudFront forwards the response to the viewer
- Author your functions in one AWS Region (us-east-1), then CloudFront replicates to its locations

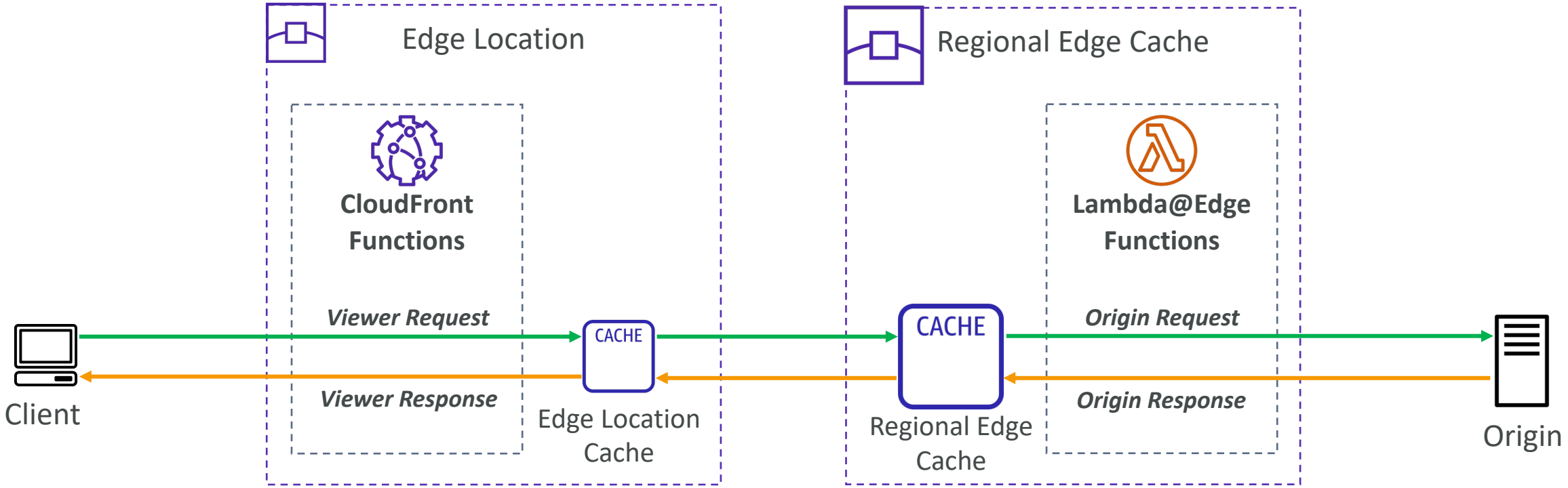



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# CloudFront Functions with Lambda@Edge

**CloudFront Functions** and **Lambda@Edge** can be used together



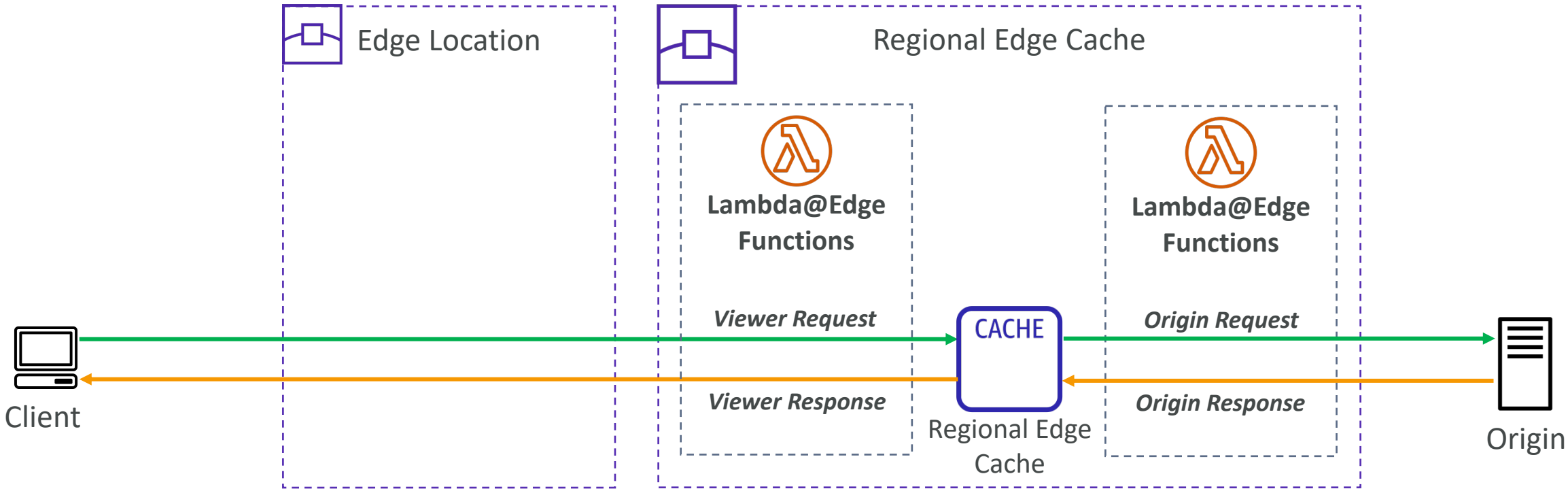

**NOTE: You can't combine CloudFront Functions and Lambda@Edge in viewer events (viewer request & viewer response)**

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# Using Lambda@Edge Only

**Use when you need some of the capabilities of Lambda@Edge that aren't available with CloudFront Functions (e.g., longer execution time, network access, …)**




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# CloudFront Functions vs. Lambda@Edge

| | CloudFront Functions | Lambda@Edge |
|---|---|---|
| Runtime Support | JavaScript | Node.js, Python |
| Execution Location | Edge Locations | Regional Edge Caches |
| CloudFront Triggers | - Viewer Request/Response | - Viewer Request/Response<br>- Origin Request/Response |
| Isolation | Process-based | VM-based |
| Max. Execution Time | < 1 ms | - 5 seconds (viewers triggers)<br>- 30 seconds (origin triggers) |
| Max. Memory | 2 MB | - 128 MB (viewer triggers)<br>- 10 GB (origin triggers) |
| Total Package Size | 10 KB | - 1 MB (viewer triggers)<br>- 50 MB (origin t |
| Network Access, File System Access | No | Yes |
| Access to the Request Body | No | Yes |
| Pricing | Free tier available, 1/6$^{th}$ price of @Edge | No free tier, charged per request & duration |

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# CloudFront Functions vs. Lambda@Edge – Use Cases

## CloudFront Functions

- Cache key normalization
  - Transform request attributes (headers, cookies, query strings, URL) to create an optimal Cache Key
- Header manipulation
  - Insert/modify/delete HTTP headers in the request or response
- URL rewrites or redirects
- Request authentication & authorization
  - Create and validate user-generated tokens (e.g., JWT) to allow/deny requests

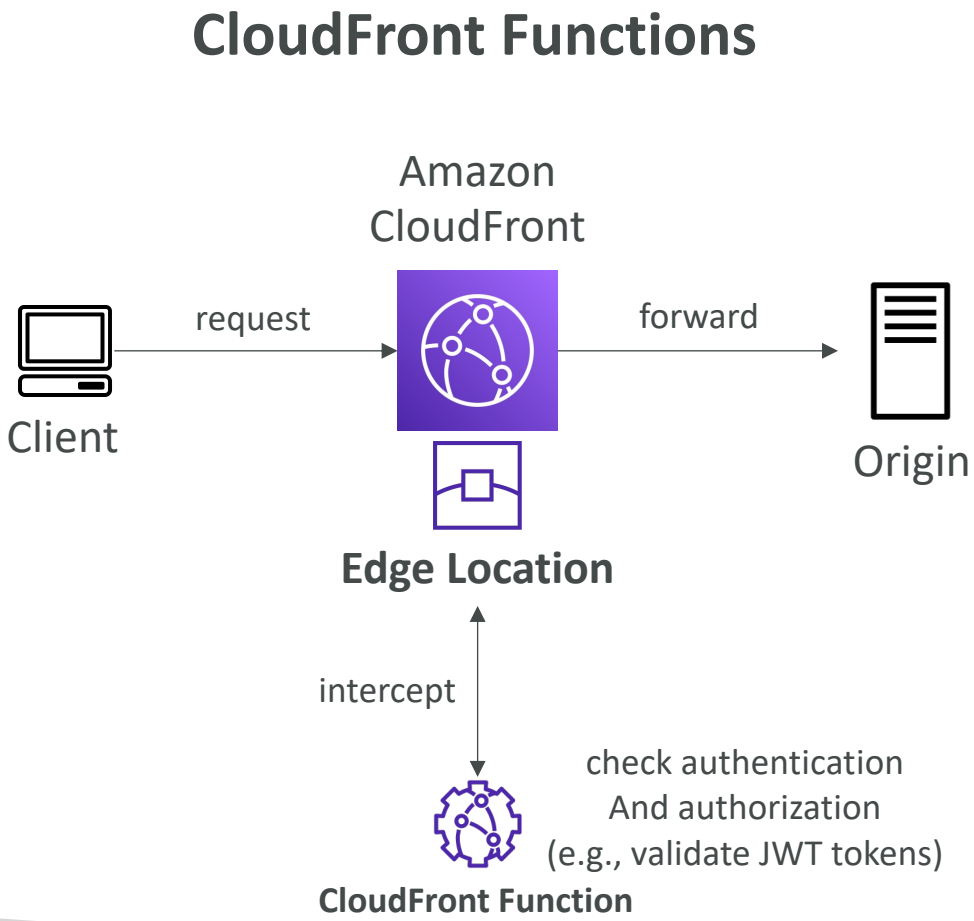
## Lambda@Edge

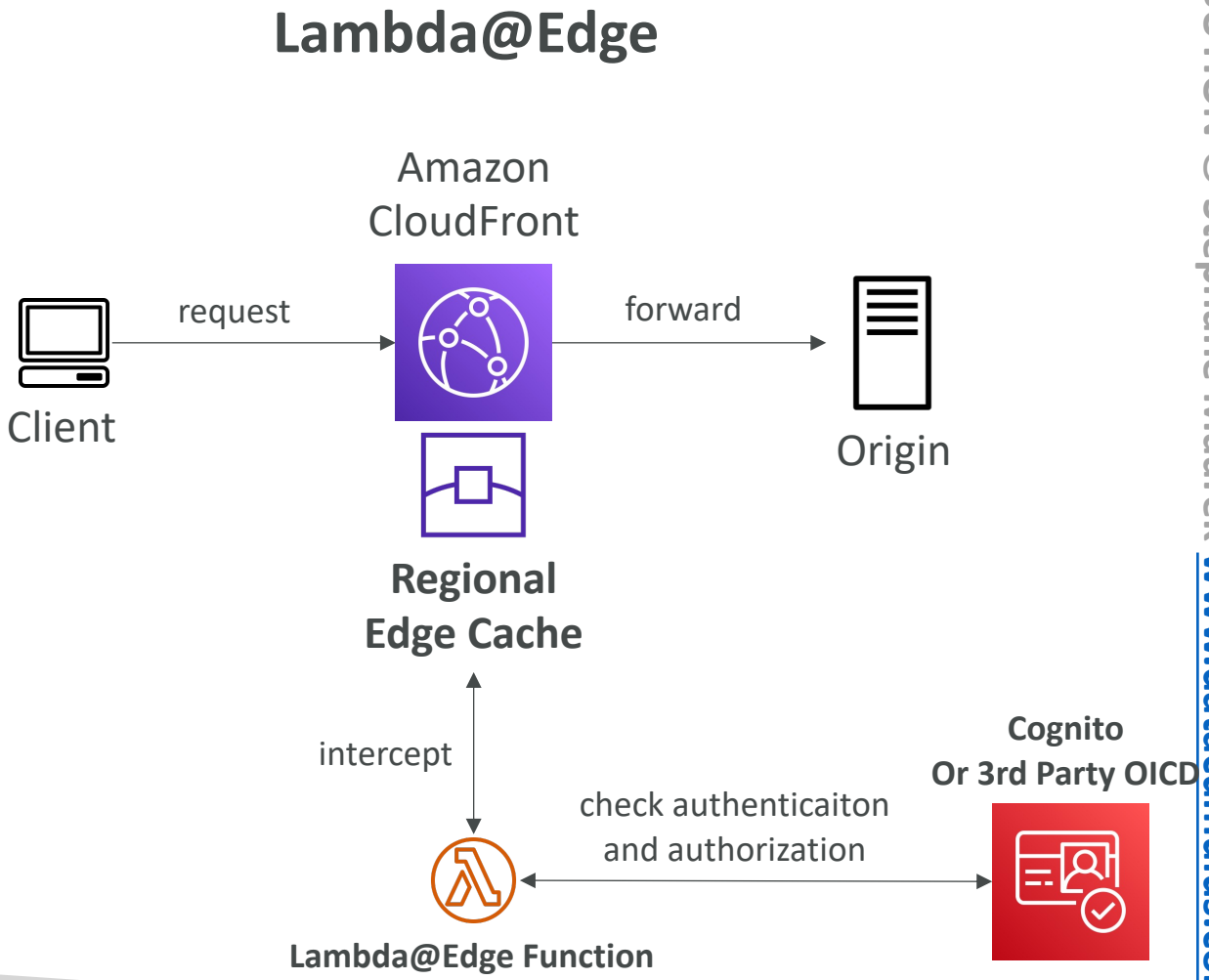
- Longer execution time (several ms)
- Adjustable CPU or memory
- Your code depends on a 3rd libraries (e.g., AWS SDK to access other AWS services)
- Network access to use external services for processing
- File system access or access to the body of HTTP requests

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# CloudFront Functions vs. Lambda@Edge – Authentication and Authorization

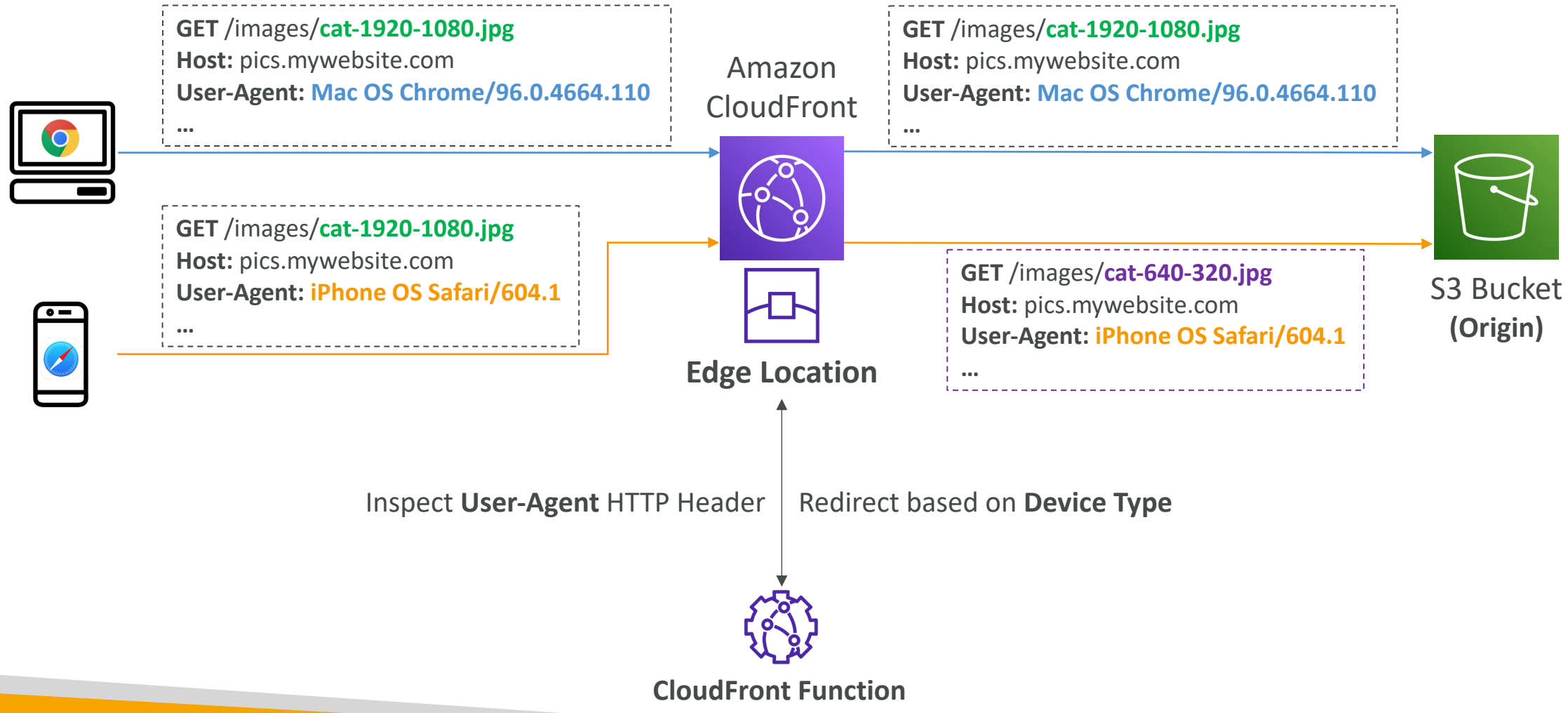



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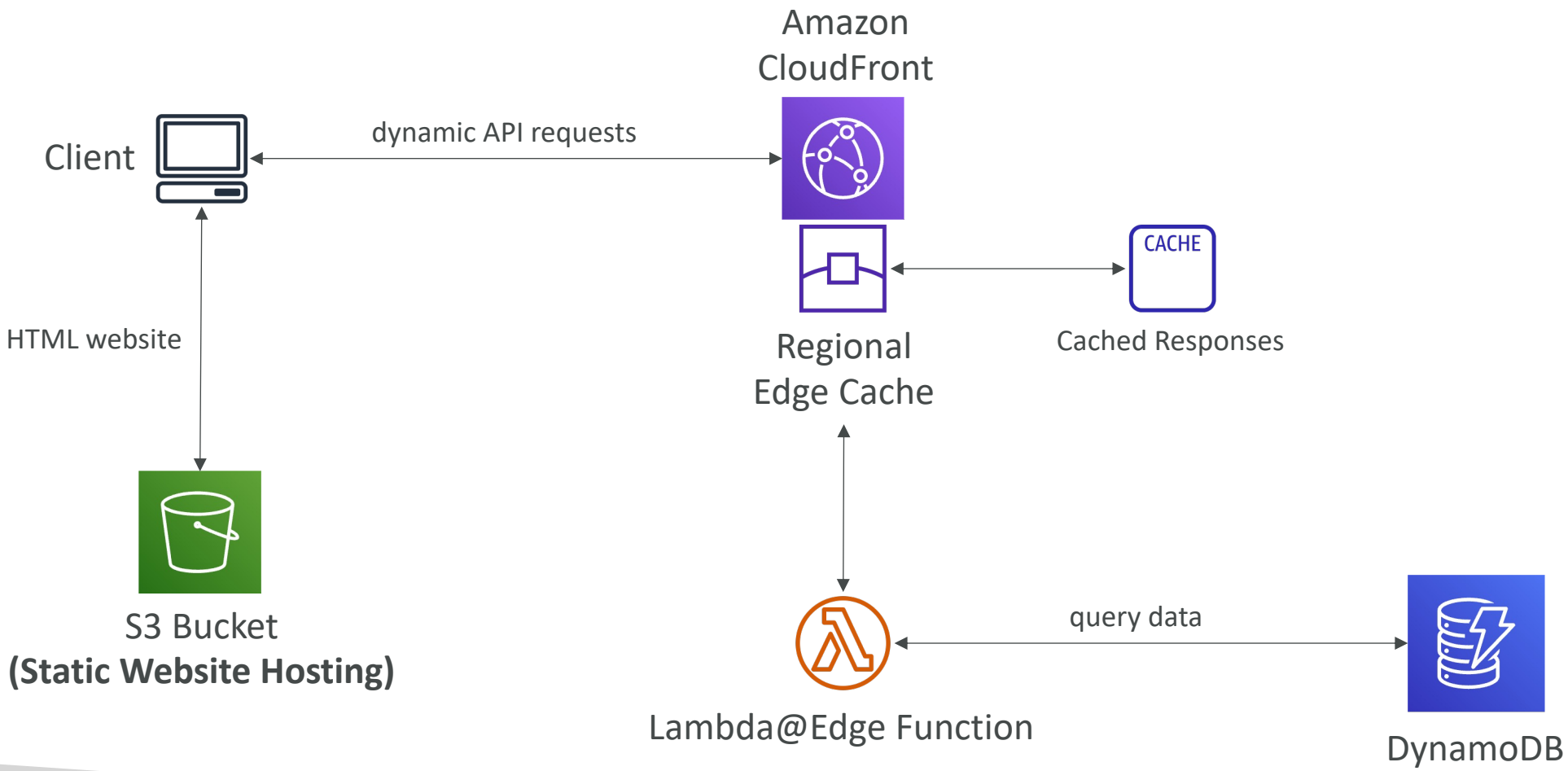
# CloudFront Functions: Loading Content based on User-Agent




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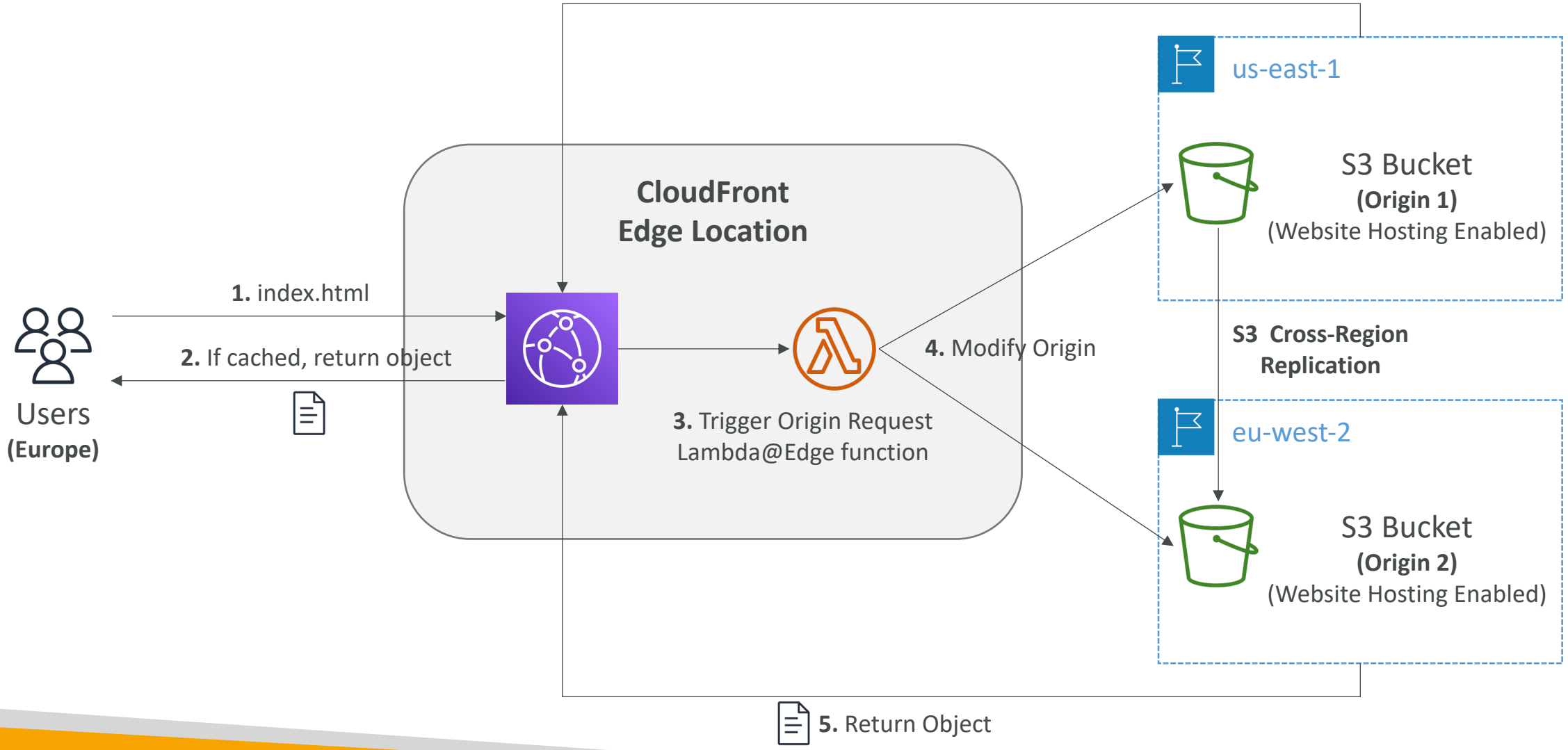
# Lambda@Edge – Global Application




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# Lambda@Edge – Route to Different Origin




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# Amazon ElastiCache Overview



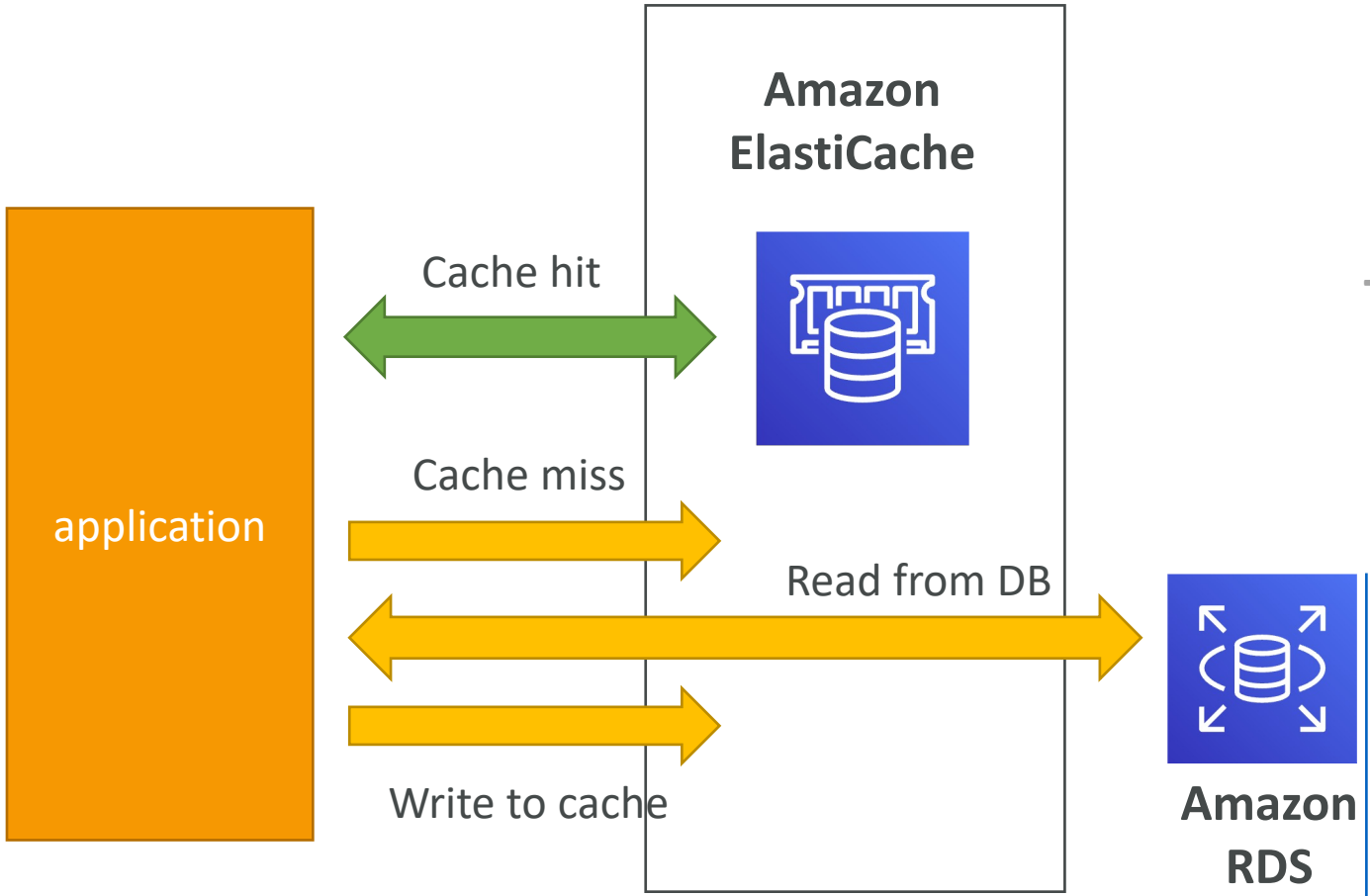
- The same way RDS is to get managed Relational Databases…
- ElastiCache is to get managed Redis or Memcached
- Caches are in-memory databases with really high performance, low latency
- Helps reduce load off of databases for read intensive workloads
- Helps make your application stateless
- AWS takes care of OS maintenance / patching, optimizations, setup, configuration, monitoring, failure recovery and backups
- Using ElastiCache involves heavy application code changes

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# ElastiCache Solution Architecture - DB Cache

- Applications queries ElastiCache, if not available, get from RDS and store in ElastiCache.
- Helps relieve load in RDS
- Cache must have an invalidation strategy to make sure only the most current data is used in there.

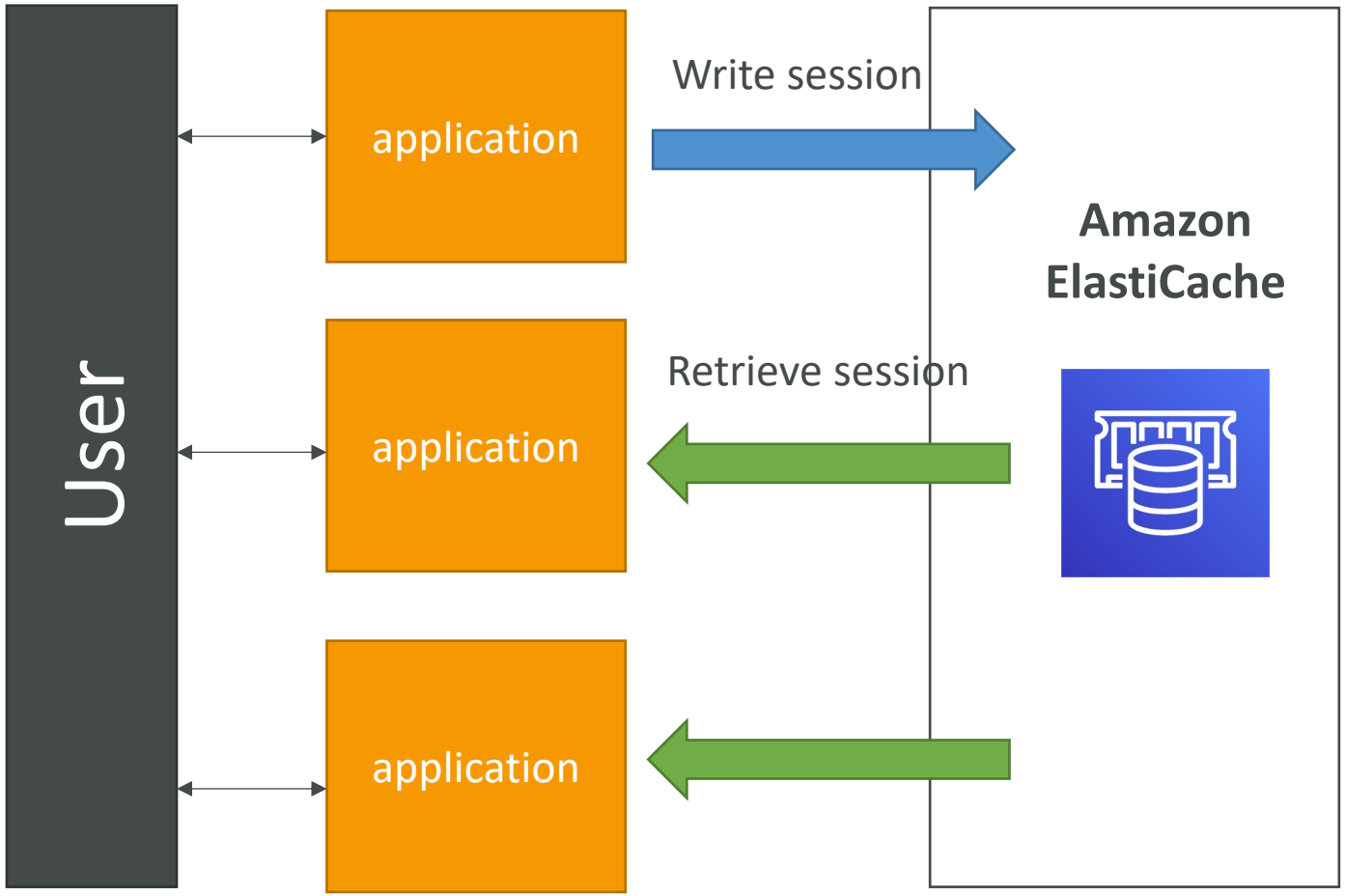



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# ElastiCache
# Solution Architecture – User Session Store

- User logs into any of the application
- The application writes the session data into ElastiCache
- The user hits another instance of our application
- The instance retrieves the data and the user is already logged in



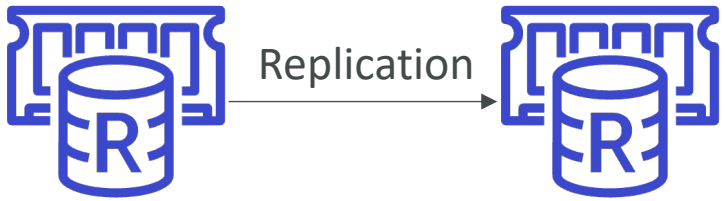

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# ElastiCache – Redis vs Memcached

## REDIS

- **Multi AZ** with Auto-Failover
- **Read Replicas** to scale reads and have **high availability**
- Persistent, Data Durability: Append Only File (AOF), **backup and restore features**




## MEMCACHED

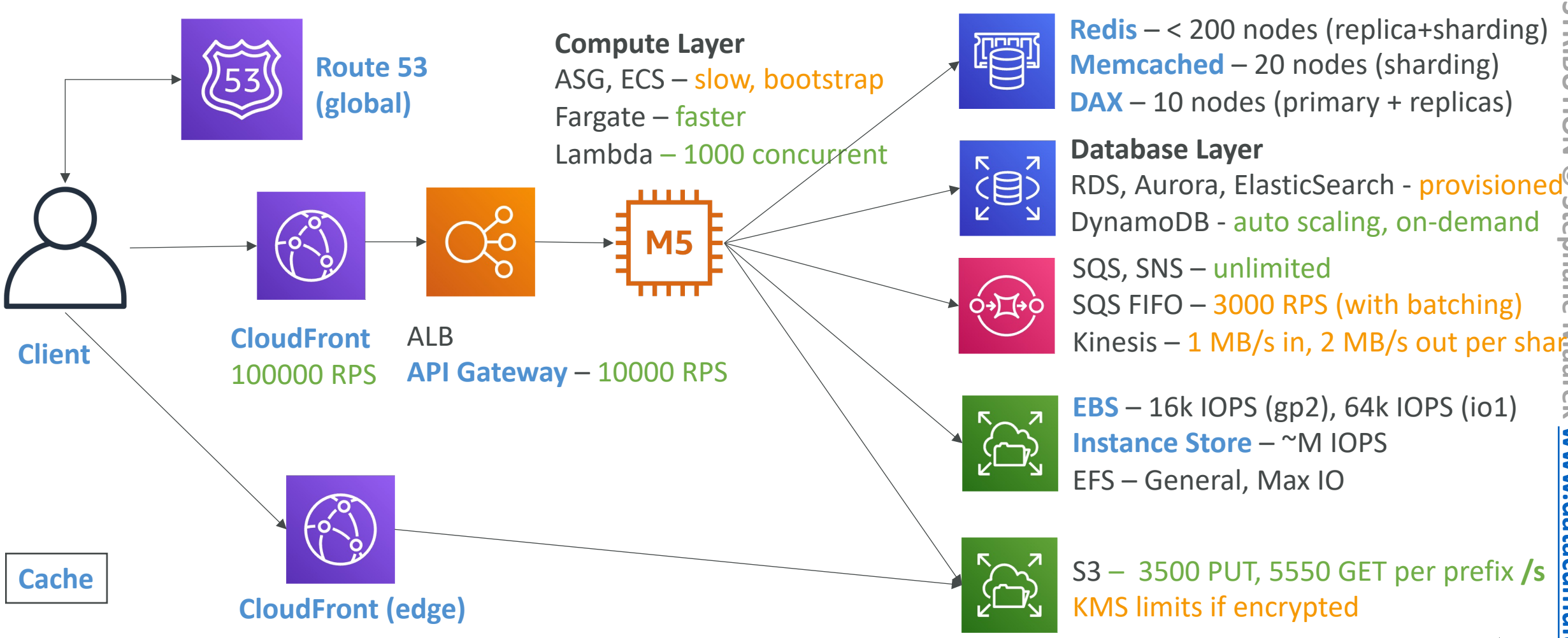
- Multi-node for partitioning of data (sharding)
- Non persistent
- Backup and restore (Serverless)
- Multi-threaded architecture




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# Handling Extreme Rates




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